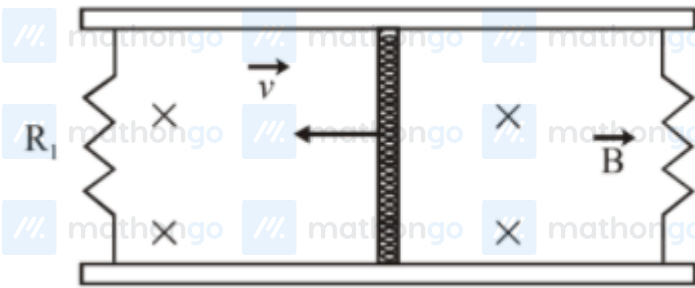


Q1: 16 March (Shift 1) - Single Correct

A conducting bar of length L is free to slide on two parallel conducting rails as shown in the figure



Two resistors R_1 and R_2 are connected across the ends of the rails. There is a uniform magnetic field $\vec{B}$ pointing into the page. An external agent pulls the bar to the left at a constant speed v . The correct statement about the directions of induced currents I_1 and I_2 flowing through R_1 and R_2 respectively is:

- (1) Both I_1 and I_2 are in anticlockwise direction
- (2) Both I_1 and I_2 are in clockwise direction
- (3) I_1 is in clockwise direction and I_2 is in anticlockwise direction
- (4) I_1 is in anticlockwise direction and I_2 is in clockwise direction

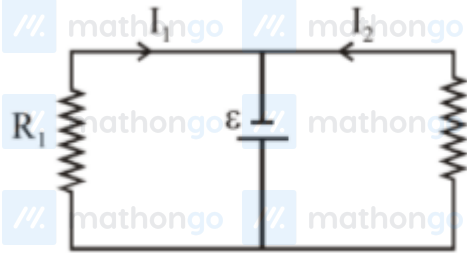
Q2: 17 March (Shift 1) - Numerical

If 2.5×10^{-6} N average force is exerted by a light wave on a non-reflecting surface of 30 cm^2 area during 40 minutes of time span, the energy flux of light just before it falls on the surface is _____ W/cm^2
(Round off to the Nearest Integer)

(Assume complete absorption and normal incidence conditions are there)

Answer Key**Q1 (3)****Q2 (25)**

Q1 (3)



Q2 (25)

$$F = \frac{IA}{C}$$

$$I = \frac{FC}{A} = \frac{2.5 \times 10^{-6} \times 3 \times 10^8}{30} = 25 \text{ W/cm}^2$$